NAQIB ZUHAIR AL-HUDRI

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PROFESSIONAL SUMMARY

Informatics Engineering undergraduate at Politeknik Negeri Jakarta and participant in the Asah AI Engineering Cohort by Dicoding and GoTo. Hands-on experience developing full-stack applications, AI-integrated software solutions, and database-driven systems. Combines technical project experience with organizational leadership, event management, stakeholder engagement, and public communication. Interested in software engineering and applied artificial intelligence, with a focus on building practical digital solutions.

EDUCATION

Politeknik Negeri Jakarta (PNJ) – Depok, Indonesia **August 2024 – September 2028 (Expected)**
Applied Bachelor's Degree (D4) in Informatics Engineering

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, PHP, C++, Java

Frontend Development: React, Vite, Tailwind CSS

Backend Development & APIs: Node.js, Express.js, Laravel, CodeIgniter 4, REST API Development

Databases & ORM: PostgreSQL, MySQL, Prisma ORM

Artificial Intelligence & Machine Learning: TensorFlow, Keras, Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Generative AI, Prompt Engineering, LLM Tool/Function Calling.

Networking: IPv4/IPv6, Subnetting, VLAN, OSPF, STP, WLAN, EtherChannel, Cisco Networking
Development Tools & Testing: Git, GitHub, Cisco Packet Tracer, Katalon Studio and Apache JMeter

PROJECTS & TECHNICAL EXPERIENCE

EduInsight – Education Data Intelligence Platform **October 2026 – November 2026**
Web In Action (WIA) 2026

- Contribute to the development of an education-focused digital platform addressing regional disparities in educational infrastructure.
- Explore open government data to support data-driven decision-making and improve public transparency in the education sector.
- Develop a solution concept that connects publicly available educational information with actionable insights for stakeholders.

Technology Stack: React, Vite, Tailwind CSS, JavaScript.

Rice Image Classification CNN Model **October 2026**
Personal / Dicoding Project

- Developed and trained a Convolutional Neural Network (CNN) using TensorFlow and Keras to classify multi-class rice grain images.
- Applied dataset preprocessing, data augmentation, normalization, and hyperparameter tuning to improve model performance.
- Achieved 98.15% test accuracy on the evaluation dataset, with a test loss of 0.0568.
- Exported the trained model into TensorFlow SavedModel, TensorFlow Lite, and TensorFlow.js formats for potential deployment across different platforms.

Technology Stack: Python, TensorFlow, Keras, TF-Lite, TensorFlow.js, NumPy.

Personal Finance Tracker & AI-Powered Financial Assistant **March 2026 – July 2026**
Personal Project

- Developed a personal finance management application featuring natural-language expense logging and AI-assisted financial interactions.
- Built a WhatsApp integration using the Baileys library and a WebSocket-based service, enabling users to record expenses through chat messages.
- Implemented multimodal input processing for receipt images and voice notes, including an OCR-based receipt-processing workflow.
- Designed an AI tool-calling engine with more than 15 tool schemas to execute structured database

operations, including CRUD, payday allocation, and budget monitoring.
- Implemented a multi-provider LLM fallback mechanism across Gemini, Cerebras, Groq, and OpenRouter to improve resilience against provider availability and rate-limit constraints.

Technology Stack: React 19, TypeScript 5.7, Vite 6, Tailwind CSS 4, Node.js 22, Express.js, Prisma ORM, PostgreSQL, Supabase, Neon, Zod, Baileys, and LLM APIs.

RuanginPNJ – Integrated Room Reservation & SQA System **September 2025 – December 2025**
Project-Based Learning (PBL)

- Contributed to an integrated room reservation system designed to support institutional room and library management.
- Performed assigned module-level software quality assurance activities, including manual testing, structural code analysis, automated UI testing with Katalon Studio, and performance testing with Apache JMeter.
- Developed administrative dashboard features and responsive UI components using Tailwind CSS v4.
- Collaborated with a five-member development team using Git for version control.

Technology Stack: Tailwind CSS v4, Katalon Studio, Apache JMeter, Git, GitHub.

WORK EXPERIENCE

Arabic Language Instructor (Insya & Tarjamah) **July 2023 – June 2024**
Pesantren Modern Daarul Uluum Lido – Bogor, Indonesia

- Taught Arabic Composition (Insya) and Translation (Tarjamah) to 150+ 8th-grade students across 5 active classes, conducting 10 teaching sessions per week.
- Designed customized weekly lesson plans, managed classroom dynamics, and applied active learning methodologies aligned with departmental curriculum standards.
- Authored comprehensive mid-term and final examination materials, evaluated student academic performance, and delivered structured progress reports.

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

AI Engineering Cohort – Asah **August 2026 – Present**
Led by Dicoding and GoTo

- Participating in a structured AI Engineering development program covering software engineering foundations, Python programming, version control, and prompt engineering.
- Developing practical competencies through structured learning activities and project-based assignments.

Belajar Fundamental Deep Learning **October 2026**
Dicoding | 110 Learning Hours

- Completed a structured curriculum covering neural networks using TensorFlow and Keras.
- Studied Computer Vision, Natural Language Processing, RNN/LSTM architectures, time-series forecasting, recommendation systems, and Generative AI.

Belajar Machine Learning untuk Pemula **October 2026**
Dicoding | 90 Learning Hours

- Completed foundational machine learning training covering supervised and unsupervised learning.
- Studied classification, regression, clustering, feature engineering, hyperparameter tuning, and model optimization.

CCNA: Switching, Routing, and Wireless Essentials **February 2026**
Cisco Networking Academy

- Completed training covering network switching, routing, and wireless technologies.
- Studied OSPF, VLANs, STP, WLANs, EtherChannel, First-Hop Redundancy Protocols (FHRP), and Layer 2 security.

CCNA: Introduction to Networks **June 2025**
Cisco Networking Academy

- Completed foundational networking training covering IPv4/IPv6 addressing, subnetting, Ethernet protocols, and router and switch configuration.

Practiced networking concepts through more than 50 Cisco Packet Tracer labs.

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

Project Officer – Grand Final Duta PNJ Batch 6 **April 2026 – August 2026**
Duta PNJ – Politeknik Negeri Jakarta, Depok, Indonesia

- Led the end-to-end planning and execution of the Duta PNJ Batch 6 recruitment process and Grand Final.

- Directed a cross-functional committee of more than 30 personnel, including 20 core committee members and 10 senior volunteers.
- Coordinated event operations and execution for approximately 300 attendees.
- Managed institutional and VIP protocols for an event featuring opening remarks by the Minister of Higher Education, Science, and Technology, Prof. Brian Yuliarto, alongside university executive leaders.
- Secured sponsorship funding and operational support through partnership outreach involving eight corporate sponsors and six national media partners.
- Established and maintained stakeholder relationships with organizations including PT Nindya, Azzura, Insta Beauty Center, Indozone, RRI, and Antara News.

President – Duta PNJ Batch 5 **August 2025 – September 2026**
Duta PNJ – Politeknik Negeri Jakarta, Depok, Indonesia

- Led a 17-member executive board of campus ambassadors, overseeing organizational strategy, work-program implementation, and internal coordination throughout the 2025–2026 leadership term.
- Directed digital branding and social media outreach initiatives across Instagram Reels and TikTok Live, contributing to organic audience growth of more than 1,000 followers.
- Served as Master of Ceremonies for more than 20 institutional and national-level events.
- Hosted high-profile events, including the FDPNI National Working Meeting, attended by the Minister of Education and directors of state polytechnics, and the PNJ–Hopu LZPU China International Partnership Launch.
- Represented the organization in institutional activities, public engagement initiatives, and external stakeholder interactions.

Logistics Lead – Open House PNJ 2026 **December 2025 – April 2026**
Politeknik Negeri Jakarta, Depok, Indonesia

- Led logistics preparation and infrastructure coordination for a two-day institutional open house welcoming nearly 1,500 prospective students from the Greater Jakarta area.
- Coordinated venue preparation, exhibition booth arrangements, and event logistics.
- Supported interactive educational activities, including the “A Day as a PNJ Student” workshops.
- Collaborated with internal teams to ensure operational readiness and a smooth visitor experience.

ADDITIONAL ORGANIZATIONAL EXPERIENCE

Flag Hoister (Pengerek Bendera) & Member **March 2021 – August 2021**
PASKIBRAKA Kabupaten Bogor – Bogor, Indonesia

- Selected through a district-wide selection process to serve as a flag hoister in the National Flag-Raising Ceremony.
- Completed intensive training and ceremonial preparation requiring discipline, physical endurance, teamwork, and responsibility.
- Demonstrated composure and coordination in a formal, high-pressure ceremonial environment.

Mentor Committee Member – EXPECTIK 2025 **November 2025**
HIMATIK PNJ

- Supported student onboarding activities and facilitated team mentoring for incoming students.

Event Committee Member – ITECHNO CUP 2025 **October 2025**
HIMATIK PNJ

- Contributed to event coordination and operational execution for a national-level technology competition.

Event Committee Member – Olimpiade Politeknik 2025 **July 2025**
BEM Politeknik Negeri Jakarta

- Supported event operations and coordination for the inter-polytechnic competition.

ACHIEVEMENTS & HONORS

Certificate of Appreciation – Director of Politeknik Negeri Jakarta **August 2026**
Politeknik Negeri Jakarta – Depok, Indonesia

- Received official institutional recognition from the Director of Politeknik Negeri Jakarta for leadership and outstanding contributions as President of Duta PNJ Batch 5.

Pramuka Garuda Award (Penegak Rank) **March 2022**
Gerakan Pramuka Kwartir Cabang Kabupaten Bogor – Bogor, Indonesia

- Awarded the highest achievement rank in Indonesian Scouting (Penegak Rank) by Kwarcab Kabupaten Bogor in recognition of leadership, personal integrity, and active community service.

Official Commendation – Regent of Bogor **August 2021**

Regional Government & Youth and Sports Agency of Bogor Regency (Dispora) – Bogor, Indonesia

- Awarded official recognition by the Regent of Bogor for exceptional discipline, leadership, and ceremonial execution as a core Flag Hoister (Pengerek Bendera) in PASKIBRAKA Kabupaten Bogor.

LANGUAGES

Indonesian: Native | **English:** Limited Working Proficiency | **Arabic:** Elementary